

Chapter 16

AI in Advocacy and Social Justice



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1 Introduction

Advocacy and social justice are critical to fostering equity, protecting human rights, and empowering marginalized communities (Selvam, 2024). Through advocacy and social justice, communities can challenge systemic inequalities and drive meaningful policy change to alleviate poverty (Desai & Solas, 2012; Jimenez et al., 2014), upholding justice for all and strengthening democratic participation (Miller, 1978). Advocacy and social justice are core to social work and guide social workers in addressing structural inequalities and empowering vulnerable populations (Dalrymple & Boylan, 2013; Watts & Hodgson, 2019). As advocates for individual and community rights, champions of public policies that promote economic and social equity, and community organizers dedicated to systemic change and human dignity, advocacy and social justice directly underpin these vital roles (Dalrymple & Boylan, 2013; Watts & Hodgson, 2019).

Artificial intelligence (AI), defined as the use of computational systems to perform tasks often associated with human intelligence, such as learning, problem-solving, and decision-making (Russell & Norvig, 2016), holds promise in facilitating change in advocacy and social justice. The depth and advances of AI enable it to robustly handle complex data types and constraints via sophisticated reasoning and analytical capabilities. This can range from text-based data using natural language processing techniques to satellite data and imagery employing remote sensing and computer vision to algorithmic economics addressing problems of equitable

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resource allocation, fairness, and systemic inequality. Thus, AI has significant potential to inform and enhance advocacy and social justice efforts by providing robust predictions, data-driven insights, and supporting evidence-based decision-making across diverse settings. These settings range from online social movements and legal contexts in judicial processes to forced migration and humanitarian assistance and the prevention of human rights violations and homelessness. AI technologies can automate and scale routine tasks, which could help relieve the excessive workload and burnout faced in social work (Lloyd et al., 2002), humanitarian assistance (Jachens et al., 2019), and legal services (Tsai et al., 2009). AI also possesses the potential to uncover valuable insights from complex data, make robust predictions to guide strategic interventions, and facilitate equitable outcomes through optimized resource allocation. Hence, this study aims to review efforts employing AI in the context of advocacy and social justice. First, we seek to summarize and categorize AI methodologies applied across a broad spectrum of applications, spanning advocacy and social justice. We define advocacy and social justice as the fight for and promotion of fairness, equity, and rights for marginalized or vulnerable groups, ensuring that individuals' civil and economic rights are recognized and protected. To remain aligned with the focus on advocacy and social justice, we primarily emphasize contexts related broadly to social equity and justice.

Second, we aim to comprehensively discuss the limitations and gaps in the application of AI in advocacy and social justice, with an emphasis on ethical considerations, including fairness, inclusivity, and strategies to ensure culturally sensitive advocacy efforts. The goal of this review is to gain a comprehensive understanding of the AI methods applied across specific domains of advocacy and social justice. These AI methods have been deployed, or have the potential to be deployed, across professions in these fields, including, but not limited to, social work. While not every social work professional will directly utilize all these AI methods, our review aims to help them understand the tools available to promote advocacy and social justice in their specialty area; empower them to leverage AI's potential in advancing these causes; and ensure they are aware of the advantages and limitations of such tools to evaluate and effectively integrate AI-driven solutions into their practice critically.

2 Background and Context

While AI has gained significant traction in recent years with the emergence of technologies such as OpenAI's ChatGPT and Google's Gemini, artificial intelligence, defined as the use of computational systems to mimic human intelligence, has been progressing steadily since its conceptualization at the "Dartmouth Workshop" in 1956 (McCarthy et al., 1956). Research applying AI to advocacy and social justice, as defined above, has been documented since the 1990s, with much of the early work relying on "rule-based" approaches, in which a set of rigorously defined rules was stored in a computer and applied to data to achieve outcomes, for example, in

housing (Barta & Kheng, 1997). This approach was dominant across many fields of applied AI research, as computers at the time lacked the processing power to support more complex methods. As such, works that apply AI tools and techniques to advocacy and social justice issues were limited, posing challenges for their application across diverse subjects in this field. At the turn of the twenty-first century, research began to advance as the rise of social media and the emergence of more powerful computing systems enabled the adoption of statistical methods to analyze different modalities of data (e.g., tabular data, text, and images), giving interdisciplinary researchers the ability and resources to analyze large volumes of diverse data—such as text, images, social networks, and tabular records—and uncover patterns or insights toward targeted issues of advocacy and social justice (Chen et al., 2008; Mochales-Palau & Moens, 2007; Fu & Chen, 2008). Such abilities to uncover insights from vast amounts of data complement the application of AI methods to social justice and advocacy issues, as context and nuance can be more effectively incorporated into analyses of systemic inequalities, public discourse, and institutional practices, thereby enabling researchers and practitioners to understand emerging challenges better, design evidence-based interventions, and strengthen advocacy strategies across diverse domains. Slightly over a decade later, the combination of far more powerful computers and advances in deep learning research (LeCun et al., 2015). The growing movement toward open-source data (Anokwa et al., 2009) enabled researchers to use AI systems to not only extract insights from complex data sources [e.g., satellite imagery (Braun et al., 2019; Redmond, 2021; McDonald et al., 2021; Mueller et al., 2021)], but also to deliver strong predictive capabilities—where models anticipate outcomes [e.g., predicting court decisions from case files (Yang et al., 2024a, 2024b)], identify trends [e.g., online networks of social movements (Suk et al., 2021; Sanghi et al., 2021; Venkatesan et al., 2021; Wahyuningroem et al., 2024)], and generate new content [e.g., chatbots used to assist advocacy groups (Pillis et al., 2024)]. We will elaborate on these in the “Results” section. These advancements, coupled with the democratization of data and tools, have made AI more accessible for social work and interdisciplinary researchers, positioning it as a powerful new “tool” to advance their work. Despite these growing promises, several dangers warrant attention. First, many open-source AI models increasingly adopted for advocacy purposes are trained on large-scale Internet data. Such data often reproduces historical and structural biases, including racist, sexist, or exclusionary content (Hofmann et al., 2024). When these models are applied directly in advocacy or social justice contexts, they risk perpetuating or even amplifying the very inequalities they are meant to address (Hofmann et al., 2024; Chanda & Banerjee, 2024). Second, just like statistical models, AI models are typically developed within specific contexts or assumptions. Transferring these systems to a different setting can lead to serious misinterpretations or inappropriate outcomes, as the assumptions baked into these methods may no longer hold (Wang et al., 2022; Chanda & Banerjee, 2024). What appears to be a reliable pattern in one environment can become a distortion in another, introducing risks for both practitioners and the communities they serve.

These risks highlight the imperative for social workers, advocates, and interdisciplinary researchers to critically evaluate how AI tools and methods are used. It is essential to ask under what scenarios these systems are appropriate and to avoid treating them as one-size-fits-all solutions. Rather than blindly applying such technologies, practitioners must remain attentive to their limitations, contexts of use, and the ethical consequences of their deployment. These risks make it essential for social workers, advocates, and interdisciplinary researchers to know when and how to use AI tools. Hence, this chapter aims to highlight both the opportunities and limitations of AI in advocacy and social justice by reviewing existing methods, applications, and future directions to inform practitioners, policymakers, and researchers on how these tools can be responsibly leveraged to advance equity, protect human rights, and support vulnerable populations.

3 Methods

3.1 Study Selection Criteria

3.1.1 Inclusion Criteria

Studies that met all of the following criteria were included in the review: (1) Analytical approach: AI; (2) Outcomes: social advocacy and social justice; (3) Time window of search: peer-reviewed articles published from the inception of a bibliographic database until February 2025; (4) Language: English; and (5) Article type: peer-reviewed journal publications, book chapters, or peer-reviewed conference proceedings. Regarding article type (5), note that, although it is common practice to exclude conference proceedings in social science-oriented reviews, we opted to include them because computer science research is frequently published in conference proceedings rather than journal articles. Furthermore, given that “AI for Social Good” has emerged as a prevalent theme across numerous computer science conferences, incorporating conference proceedings ensures comprehensive coverage of relevant, state-of-the-art contributions from the interdisciplinary literature.

3.1.2 Exclusion Criteria

Studies that met any of the following criteria were excluded from the review: (1) letters to the editor, dissertations, or case reports; (2) preprint articles; and (3) studies focusing on advocacy and justice within adjacent domains not directly related to social advocacy and justice. (Refer to the later chapter for climate advocacy and justice and prior chapters for works concerning healthcare/clinical advocacy and justice.)

3.2 *Search Strategy*

3.2.1 *Databases*

A keyword search was performed across seven electronic bibliographic databases: PubMed, Web of Science (WoS), EBSCO, Scopus, Cochrane Library, Association for Computing Machinery (ACM) Digital Library (DL), and Institute of Electrical and Electronics Engineers (IEEE) Xplore, reflecting a balanced selection of both computer science conference-based publications and non-computer science journal-based sources to comprehensively capture the diverse publication avenues across this interdisciplinary research area.

3.2.2 *Keyword Search*

The search strategy included comprehensive keyword combinations from two primary conceptual groups: (1) Artificial Intelligence (AI) and related methodological terms and (2) advocacy and social justice. For the AI domain, we included broad terms such as “Artificial Intelligence” and “Machine Learning”; specific subdisciplines, including “Natural Language Processing,” “Computer Vision,” and “Deep Learning”; related methodological concepts such as “Data Mining,” including “Text Mining” and “Social Media Mining”; and “Big Data.” For the Advocacy and Social Justice domain, we first considered terms explicitly related to advocacy, including “Advocacy,” “Social Movement,” and “Marginalized,” and terms directly related to social justice, including “Social Justice,” “Racial Justice,” “Racial Movement,” “Human Rights,” and “Civil Rights.” All keywords in PubMed were searched with the “[Title/Abstract]” tag, which is processed using Automatic Term Mapping. The search function TS = Topic was used in Web of Science, EBSCO, Scopus, Cochrane Library, ACM DL, and IEEE Xplore, which searches for topic terms across the title, abstract, and keywords fields. The search algorithm for spanning these five electronic bibliographic databases is reported in the Supplemental Materials.

4 *Results*

4.1 *Literature Search*

Figure 16.1 presents the PRISMA flow diagram. We identified 7715 articles through keyword searches, including 2505 from Scopus, 1963 from EBSCO, 1426 from Web of Science, 960 from IEEE Xplore, 443 from PubMed, 399 from the ACM Digital Library, and 19 from the Cochrane Library. After removing 2516 duplicate articles and 2 retracted articles, 5197 articles were retained for screening.

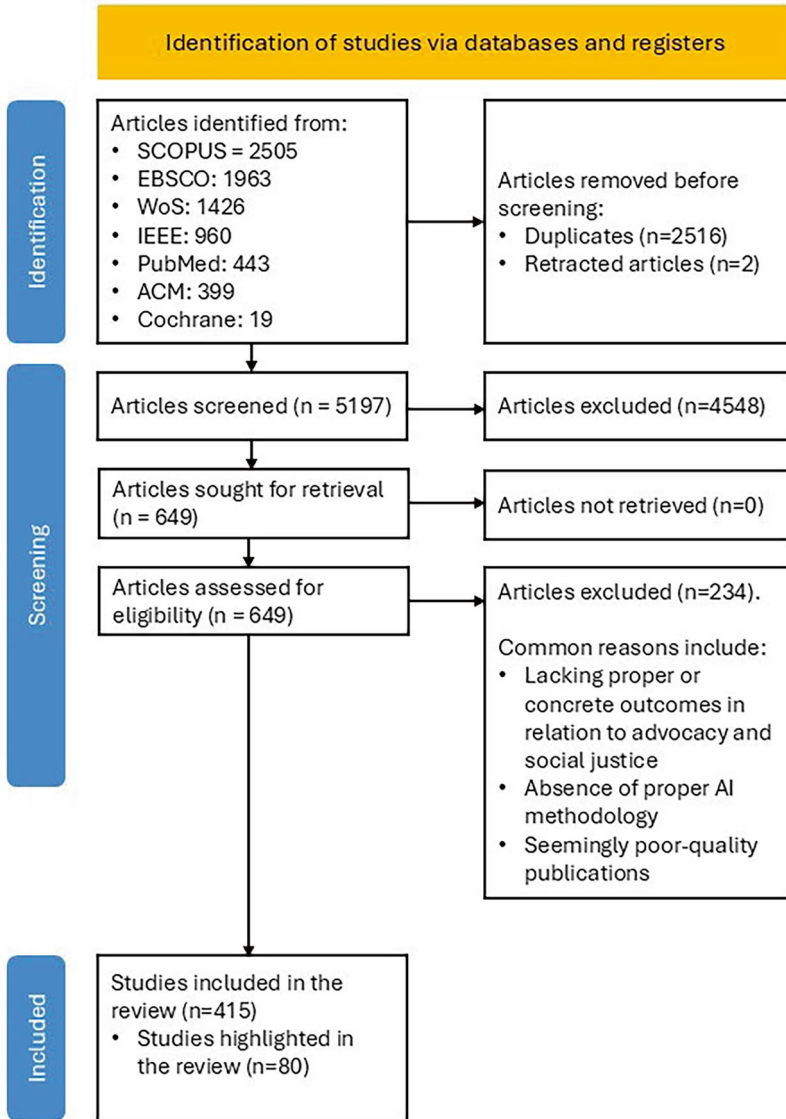


Fig. 16.1 PRISMA diagram

These articles underwent title and abstract screening, thereby eliminating 4548 articles. Common reasons for the exclusion of these articles included (1) no relation to social justice or advocacy (e.g., articles where these terms were used passively), (2) papers that were purely computational in nature, (3) articles concerning the implications of AI in social justice and advocacy as opposed to its application, (4) articles concerning other aspects of advocacy and justice (refer to prior chapters for

works concerning healthcare advocacy and the later chapter for climate advocacy), and (5) articles for which the main text was unavailable in English. Subsequently, 649 articles were selected for full-text reviews, of which 234 were excluded for common reasons including, but not limited to, (1) lacking proper or concrete outcomes in relation to advocacy and social justice, (2) the absence or lack of proper AI methodology (e.g., using basic regression techniques as part of the methodology while using the terms “AI” and “statistical analysis” interchangeably), and (3) seemingly poor quality publications. As a result, 415 articles were included in this review.

Due to the sheer number of papers on this topic, we opt to highlight representative and prominent examples within each methodology cluster with respect to their relevant applications, as outlined in the subsequent subsections below. In total, we highlighted 80 studies.

4.2 Study Characteristics

Figure 16.2 shows the trends of the number of papers surrounding AI for advocacy and social justice published over the past 15 years, reflecting a steady exponential increase. Specifically, over the past 10 years: 12 were published in 2016, 18 were

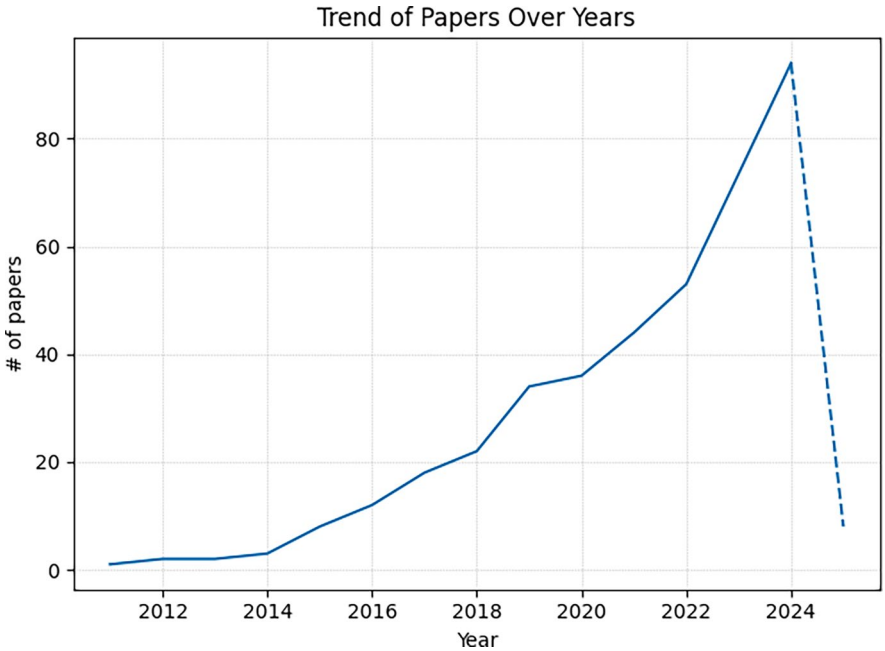


Fig. 16.2 15-year trend of the number of papers surrounding AI for Advocacy and Social Justice published

published in 2017, 22 were published in 2018, 34 were published in 2019, 36 were published in 2020, 44 were published in 2021, 53 were published in 2022, 71 were published in 2023, 94 were published in 2024, and 8 papers have been published thus far in 2025.

Of the papers filtered through our inclusion and exclusion criteria, 217 were conference proceedings, 197 were journal articles, and 1 was a book chapter, reflecting a balance between computer science-focused publication avenues, which typically emphasize computational innovations and methodological advancements, and journal articles that are the preferred publication avenues among social scientists and social work academics.

4.3 Overview of Distinct AI Methods Used to Advance Advocacy and Social Justice

Among the papers selected for review, we grouped them into distinct categories based on AI methodologies and their respective applications toward advancing advocacy and social justice: (1) Web and social media mining applications for advocacy (n = 204), (2) NLP applications assisting legal efforts in pursuing social justice (n = 79), (3) Computer vision and remote sensing applications solving humanitarian challenges (n = 33), (4) Virtual reality and chatbots supporting advocates and stakeholders (n = 11), and (5) others (n = 88). We illustrate this categorization as a conceptual framework in Fig. 16.3.

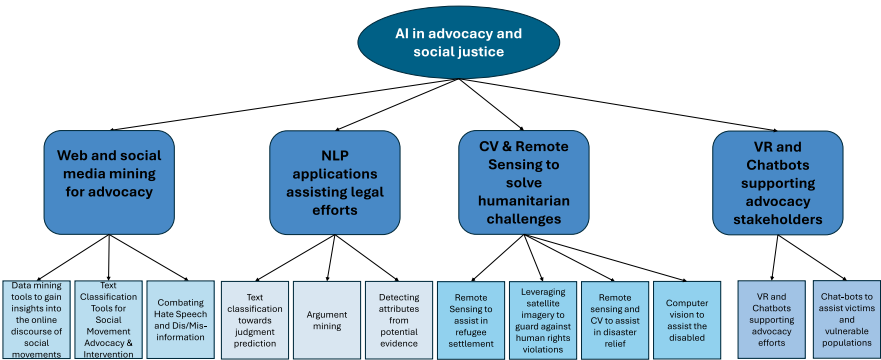


Fig. 16.3 Distinct buckets of AI methodologies and their respective applications

4.3.1 Web and Social Media Mining Applications for Advocacy

Social media has emerged as a prominent platform for organizing, amplifying, or complementing modern social movements. It raises awareness, highlights lived experiences, and expresses diverse opinions on various movements. Platforms like X and Reddit provide avenues for individuals to voice their opinions or experiences on a massive scale, uncovering diverse perspectives that traditional surveys, often constrained by limited sample sizes, might overlook (Alba & An, 2023). Additionally, social media's global connectivity allows researchers to better understand complex social phenomena through methods such as community detection, network analysis, and social network visualization, thereby enhancing the depth and breadth of advocacy-related insights.

Advancements in data mining methods have created opportunities for researchers to apply sophisticated analytical techniques to harness insights from “big data” to inform actionable interventions in social media. This is otherwise known as “social media mining,” which is the process of collecting and analyzing user-generated content from online platforms, including social networking sites (e.g., Facebook, X), online forums (e.g., Reddit), video-sharing platforms (e.g., YouTube), and search engines (e.g., Google Trends), to identify and interpret patterns, trends, and actionable insights. Unlike some of the later-mentioned techniques, social media mining does not rely on a single method; rather, it is an interdisciplinary approach that integrates various analytical tools, including natural language processing and social network analysis, to leverage social media data effectively.

Data Mining Tools to Gain Insights into the Online Discourse of Social Movements

Data mining tools offer opportunities to amplify voices within social movements by leveraging insights from vast volumes of user-generated content, providing a scalable approach to understanding collective opinions and identifying key messages from marginalized groups.

Network analysis is a popular data-mining tool used to analyze connections across online interactions and the dynamics among individuals, groups, and key actors within social movements. By drawing on critical insights into online community structures and sources of influence, advocates can identify and target influential leaders to support online collaborative efforts and strengthen community resilience. For example, Jeon (2020) utilized network analysis to examine the interplay between narratives and entities in personal accounts of sexual violence in sports, following revelations related to Olympic gold medalist speed skater Suk-Hee Shim. Similarly, Suk et al. (2021) leveraged network analysis to understand how shared personal experiences of sexual violence and harassment on Twitter during the #MeToo movement transcended into calls for action across a range of institutions and communities. This allowed advocates to identify key discourses, actors, and themes associated with shared experiences surrounding the #MeToo movement.

Network analysis was also employed across a variety of political and democratic movements (Sanghi et al., 2021; Venkatesan et al., 2021; Wahyuningroem et al., 2024). For instance, Wahyuningroem et al. (2024) used network analysis to examine the influential role of youth participants in Indonesia's #ReformasiDikorupsi movement.

Topic modeling is another popular tool for uncovering the major themes and narratives in social movements. In summary, topic modeling helps identify prevalent themes or "topics" within vast collections of user-generated online texts, enabling advocates to distill complex volumes of online discussions into targeted, actionable insights that directly inform strategy, communication, and outreach efforts. For instance, Lee and Jang (2023) and Tong et al. (2022) applied topic modeling to identify central topics in personal narratives and broader social media discourse related to the #StopAsianHate and #BlackLivesMatter movements. Similarly, Umel (2024) used topic modeling to understand the struggles of Filipino migrants in Germany. These examples illustrate how themes derived from topic modeling provide systematic, yet insightful summaries of the dominant concerns, experiences, and perceptions expressed on social media, enabling nonprofits and advocacy groups to make informed decisions and implement strategic interventions.

Leveraging Text Classification Tools for Social Movement Advocacy and Intervention

Text classification involves leveraging machine learning tools to classify text into predefined categories. This capability helps researchers and practitioners systematically detect critical events, sensitive content, and specific behaviors across large volumes of online communication, facilitating timely interventions and targeted responses to emerging issues. For instance, Khatua et al. (2018) and Sawhney et al. (2021) employed text classification techniques to identify places (e.g., workplace, home, public spaces) and emotions linked to personal narratives of sexual violence arising from the #MeToo movement. Similarly, Subramani et al. (2019) developed text classifiers to identify the communicative strategies in online posts from victims or supporters of domestic violence, ranging from awareness to empathy to personal stories. These efforts could enable the early identification of emotional triggers and communication patterns associated with sexual and domestic violence, thus informing proactive strategies such as targeted awareness campaigns, educational programs, and community interventions designed to mitigate risk factors and foster environments that deter such violence. In another application, Stuart et al. (2020) employed text classification techniques to detect social media texts associated with traumatic grief from Chicago gang territories. The authors utilized insights gathered from in-depth interviews with neighborhood youths and observations from stakeholders (gangs, participants of violence-reduction programs, and workers of violence prevention organizations) to inform the development of their algorithms, capturing how traumatic grief is expressed on social media among residents in gang territories. These machine learning algorithms were deployed at scale to

systematically analyze online data and gather critical insights, which were subsequently used to assist community organizations and violence prevention initiatives in developing more effective tools—including trauma-informed intervention applications—for victims of gang violence. *Sentiment analysis* is another popular text classification tool used to analyze evolving public sentiments toward social movements. In another instance, Haimson (2019) used sentiment analysis on Tumblr blogs to identify sentiment patterns associated with transgender identity disclosures and social support among transgender individuals, while Swanson et al. (2023) and Sabatovych (2019) used sentiment analysis to capture the prevailing emotional attitudes toward abortion rights movements following the *Dobbs v. Jackson Women's Health Organization* ruling. These examples illustrate how sentiment analysis could help women's and LGBTQ advocates capture emotional trends and shifts in public opinion, enabling targeted interventions, outreach, and inclusive policymaking.

Combating Hate Speech and Dis/Misinformation

Detecting hate and offensive speech, particularly targeting marginalized or minority communities and individuals, remains a core component of social media mining efforts in advocacy and social justice. While the rapid rise of social media has enhanced societal connectivity, the anonymous and non-face-to-face nature of interactions in the digital realm has facilitated an increase in hateful and offensive speech. AI could therefore be used to detect and filter such hateful and offensive online texts, including women-targeted online abuse (Rathod et al., 2023) and misogynistic aggression (Li, 2022), hate speech toward LGBTQIA+ individuals (Bozhidarova et al., 2023), and racially motivated offensive speech or slurs (Gupta et al., 2023).

Similarly, social media's rapid ability to disseminate information, often with minimal accountability, has facilitated the rise of misinformation and disinformation, which can undermine trust, distort public perception, and hinder efforts to achieve equitable and just outcomes in communities. Although no unified method exists to combat hate speech and misinformation, advancements in artificial intelligence, including some of the previously discussed techniques, offer promising avenues to address these challenges. While detecting hate speech remains challenging due to evolving slang and contexts, recent advances in language models have significantly enhanced identification and response capabilities. For instance, Anandhi et al. (2024), Chit et al. (2022), and Hüsünbeyi et al. (2022) leveraged advanced text classification methods to detect hate speech online, revealing nuanced patterns of discriminatory language and enabling quicker intervention. Similarly, tools such as social network analysis (Sivasankari & Vadivu, 2022) have been deployed to combat mis/disinformation by tracing the propagation paths of false information, identifying influential sources responsible for its widespread dissemination, and thereby enabling targeted interventions to disrupt online misinformation networks effectively. More recently, Generative AI, a type of AI technology capable of producing original content, such as ChatGPT, has also been employed to combat

misinformation and disinformation (Jaidka et al., 2025) by generating fact-checking and explanatory content to debunk false claims proactively, address misconceptions, and promote digital literacy initiatives.

4.3.2 NLP Applications Assisting Legal Efforts

A variety of NLP tools are employed to predict, gain insights, and interpret legal documents and judicial processes. These methods range from forecasting judicial outcomes based on the presented arguments to leveraging historical data to recommend arguments to prosecutors and defenders. Collectively, these approaches could aid the pursuit of social justice by facilitating, assisting, and expediting legal processes, thereby striving toward accountability for human rights violations and protecting the innocent, while promoting fair and equitable trials.

Text Classification Toward Judgment Prediction

Building models capable of predicting court decisions remains a prominent text classification application in the legal domain. Typically, these models incorporate case law documents as input, including facts, legal arguments, judicial reasoning, interpretations, and verdicts, and can subsequently assist defenders and prosecutors by forecasting case outcomes based on proposed arguments and existing case files. This predictive capability enables legal professionals to prepare their cases better, develop stronger arguments, and strategize more effectively, ultimately promoting fairer and more informed judicial processes. This predictive capability may further enable proponents of social justice to achieve favorable judicial outcomes, such as Supreme Court rulings supporting civil rights, equal protection, and antidiscrimination efforts. Advancements in language models, specifically designed to process textual inputs and generate predictions with accuracy rates up to 91.8% (Yang et al., 2024a, 2024b), have enabled more reliable deployment of judgment prediction tools, significantly aiding legal professionals, including forensic and legal social workers, in effectively preparing for court cases. Such models have been built to predict judgments across a variety of countries and levels of government, ranging from the European Court of Human Rights (ECHR) to Supreme Court rulings in the United States (Yang et al., 2024a, 2024b), China (Wang et al., 2024), and India (Yang et al., 2024a, 2024b), to rulings made in local courts in Indonesia (Sari et al., 2022).

While accurate predictions offer considerable value, understanding the underlying reasons behind a model's predictions remains essential. Explainable AI (xAI) addresses this by clarifying how specific arguments or legal factors influence the model's judgment prediction outcomes. Such insights empower legal and forensic social workers to strategically assess case strengths, refine advocacy strategies, and promote greater transparency and fairness in judicial processes. Consequently, XAI is becoming increasingly valuable in legal decision-making, transitioning from

traditional black-box predictions to transparent, interpretable explanations that are critical for trustworthy forecasting in legal contexts. Several methods and frameworks, such as Abstract Dialectical Frameworks (ADFs) (Collenette et al., 2023; Mumford et al., 2023) and the ANGELIC methodology (Dupré, 2023; Mumford et al., 2023), have been developed to elucidate the complex legal reasoning underlying AI prediction models. For instance, the latter's ANGELIC methodology organizes legal reasoning hierarchically—from verdicts to issues, abstract factors, and factual base-level factors—and produces natural language, step-by-step explanations as part of its prediction process, clearly showing how the model arrives at its decisions (Dupré, 2023; Mumford et al., 2023). Beyond predicting judgments from case law documents, these advancements in language models have also enabled the identification of specific violations detailed within court documents. Such capabilities allow forensic and legal social workers to swiftly identify relevant human rights violations, expediting the pursuit of justice, facilitating timely interventions, and potentially integrating directly into judicial workflows to enhance court efficiency and accuracy. This approach has been demonstrated effectively in identifying specific articles of violation in judgments issued by the European Court of Human Rights (ECHR) (Imran et al., 2023; Kaur & Bozic, 2019; Mumford et al., 2024; Quemy & Wrembel, 2020; Santosh et al., 2024; Xu et al., 2024; Yıldırım et al., 2023), made possible due to openly available court documents and transparency in the ECHR judicial processes.

Collectively, these AI-driven methodologies for legal prediction can support social justice advocates by enabling more accurate forecasting of judicial outcomes and by strengthening the preparation of evidence-based arguments, ultimately helping advocates secure favorable court decisions that advance social justice.

Argument Mining

Argument mining is another valuable methodology that can help legal professionals identify key aspects of arguments, evidence, and reasoning in court cases. Unlike the aforementioned text classification approaches, which primarily focus on predicting court outcomes, argument mining mainly focuses on extracting, clustering, and distinguishing the characteristics of arguments that influence a verdict. These features provide valuable insights to help advocates and legal social workers enhance their strategic argumentation and decision-making in their pursuit of fair and equitable judicial outcomes. There are various methodological approaches to argument mining. One method is to train a text classifier to identify predefined argumentative characteristics in a given piece of text. This is exemplified by Habernal et al. (2024), who developed a multitask text classification model capable of identifying a variety of precise argument types, such as interpretive methods, principles of proportionality, and institutional arguments, that are associated with distinct judicial actors and outcomes. These classifications could assist justice advocates and legal social workers in strategically identifying and strengthening their arguments in their pursuit of legal justice.

Another approach is to employ unsupervised clustering methods that naturally group sentences into argumentative categories based on their similarity. Unlike the above-mentioned approach, which classifies a piece of text into a predefined category, unsupervised methods cluster text with similar argumentative patterns “naturally” into groups, allowing legal social workers and advocates to flexibly explore emerging or previously unidentified argumentative themes without being constrained by the predefined classifications used in previous methods. For instance, Poudyal et al. (2019) developed unsupervised clustering algorithms to distinguish legal arguments related to human rights violations in legal documents. Specifically, they exploited the “Fuzzy” algorithms to account for the fact that a single sentence could encompass multiple arguments, thereby effectively addressing overlapping argumentative content often seen in legal contexts, enabling social workers and legal advocates to better recognize and navigate complex legal reasoning patterns in court cases.

Developing comprehensive yet accessible argument-mining tools is crucial for enabling legal social workers and advocates to easily deploy them across diverse legal scenarios. This is exemplified by Poudyal (2015), who developed a comprehensive argument-mining tool that consists of an (1) Argument Element Identifier (AEI), which uses machine learning algorithms to distinguish argumentative from non-argumentative sentences; (2) Argument Builder (AB), which employs the aforementioned unsupervised algorithms to group similar argumentative sentences into related clusters; and (3) Argument Structurer (AS), which uses word detection and contextual analysis to differentiate between premises and conclusions within argumentative clusters. Such comprehensive argument-mining tools create an integrated pipeline to streamline legal analysis and strategically identify key argumentative characteristics, thereby assisting legal social workers in building stronger cases in a time-efficient manner in their pursuit of social justice.

Detecting Attributes from Potential Evidence

Leveraging AI models to detect critical indicators, such as specific elements of torture, trauma, and abuse within extensive text-based evidence documents, can assist prosecutors, legal professionals, or forensic social workers in gathering relevant evidence. Such techniques not only streamline the evidence review process but also enhance accountability in the pursuit of justice. For instance, Gokhale and Fasli (2017), Kihlman and Fasli (2019), and Gokhale and Fasli (2018) developed text classification models that can detect and identify torture mechanisms such as hooding, blackened goggles, plastic blindfold, and sight deprivation by other means, which are types of sensory deprivation, from testimonies and survivor stories. Similarly, Schirmer et al. (2024) developed GENTRAC, a language model that detects trauma from genocide and mass atrocity court transcripts. Such models could henceforth amplify the voices of marginalized victims, empower survivors by validating their experiences, and enable advocates and legal social workers to pursue accountability and systemic change through robust, evidence-based advocacy.

Additionally, models can also be deployed to assist judges, legal professionals, and forensic social workers in evaluating the reliability and credibility of witness testimony. For instance, Gustafsson et al. (2025) developed machine learning models capable of predicting the accuracy of eyewitness statements based on 94 distinct acoustic features extracted from audio recordings. This demonstrates the potential of using AI to support the legal system in evaluating witness credibility, thereby contributing to fairer judicial outcomes and processes in the pursuit of social justice.

4.3.3 Computer Vision and Remote Sensing to Solve Humanitarian Challenges

Every second, millions of images are captured worldwide, ranging from satellite imagery to ground-based photographs. Each image inherently “contains a thousand words,” offering rich information that can address some of the most pressing humanitarian challenges of our time, including refugee settlement planning, detection of human rights violations, disaster relief coordination, and accessibility for individuals with disabilities. However, manually extracting actionable information from such vast amounts of dense imagery can be laborious and time-consuming. For instance, analyzing satellite images manually after natural disasters to identify damaged infrastructure or optimal relief routes can significantly delay response efforts, potentially endangering lives. In critical contexts where human rights and lives hang in the balance, timely intervention is paramount, as delays can lead to increased suffering, prolonged displacement, and preventable loss of life. Therefore, computer vision, a field within AI focused on enabling machines to interpret and predict visual information from images or videos, and remote sensing, the discipline involving the acquisition of information remotely through satellite or aerial imagery, can provide invaluable assistance by enabling the rapid extraction of crucial insights to enhance the effectiveness and timeliness of humanitarian responses and interventions by authorities, NGOs, and advocacy groups dedicated to achieving social justice.

Remote Sensing to Assist in Refugee Settlement

Satellite imagery provides critical information to monitor refugee movements, detect and map settlement expansion, assess environmental impacts, and identify specific infrastructure within refugee or internally displaced people (IDP) settlements. Remote sensing techniques, which analyze and interpret satellite imagery to extract meaningful insights, play a crucial role in translating raw satellite data into actionable information. Such information can greatly benefit humanitarian workers by enhancing their situational awareness, informing efficient resource allocation, and ultimately improving response times and effectiveness during refugee crises.

For instance, Braun et al. (2019) used remote sensing to assess refugee camp expansion and associated environmental changes with the refugee crisis in

Kutupalong, Bangladesh, in lieu of the Rohingya genocide from neighboring Myanmar. The results showed a significant expansion of the refugee camp area, increasing from approximately 36 hectares in early 2016 to over 1850 hectares by early 2019, and a corresponding vegetation loss of about 1500 hectares, highlighting severe environmental impacts associated with rapid refugee influx. These insights can help humanitarian and social justice advocates build stronger cases for international support and peace advocacy, as well as enable more informed planning and resource allocation to manage settlement expansions and associated humanitarian needs effectively. Similarly, Quinn et al. (2018) employed computer vision techniques on high-resolution imagery from 13 refugee/IDP settlements in South Sudan, Iraq, and Nigeria to quantify the distinct structural types within these settlements. Automating this otherwise labor-intensive manual task can assist humanitarian social workers in settlement monitoring, resource and infrastructure planning, and vulnerability and needs assessment.

Additionally, Moser et al. (2022) demonstrated the effectiveness, robustness, and accuracy of advanced computer vision models for identifying small vessels in satellite imagery, with the ultimate aim of leveraging these techniques to enable NGOs and the public to detect refugee boats more effectively. These advancements strengthen humanitarian responses, support timely rescue interventions, and safeguard refugees in vulnerable situations at sea, equipping social workers with critical insights to advocate effectively for humane migration policies and greater international accountability.

Leveraging Satellite Imagery to Guard Against Human Rights Violations

Satellite imagery provides essential data for detecting, classifying, and documenting human rights violations, such as forced displacement, violent conflict, and forced labor. Leveraging remote sensing and computer vision techniques, AI can rapidly deliver predictive insights to humanitarian workers, facilitating urgent, evidence-based interventions and accountability. Such timely intelligence supports social justice by ensuring transparent monitoring, equitable protection, and swift responses to human rights abuses.

Redmond (2021) used remote sensing on satellite imagery to monitor violent conflict targeting civilians during the Rohingya genocide by assessing the destruction of settlements in Rakhine State. Specifically, through semantic segmentation, where each pixel in a captured satellite image was classified as either damaged or undamaged, Redmond was able to quantify and geo-reference the precise amount of damage in Rakhine State at the time the image was captured. By scaling this across satellite imagery captured at different times spanning the genocide, humanitarian workers could assess the scale of civilian displacement and identify affected areas, enabling social and humanitarian workers to direct timely assistance and targeted interventions to those most in need.

Similarly, McDonald et al. (2021) used satellite-based vessel-tracking data, combined with machine learning, to identify globally high-risk fishing vessels.

Specifically, they analyzed data from over 16,000 industrial fishing vessels to detect behavioral patterns indicative of forced labor, including extended time at sea, infrequent port visits, and transmission gaps. Their approach provides targeted insights that can assist humanitarian and social workers in identifying vulnerable populations, guiding resource allocation, planning effective interventions, and advocating effectively for the protection and rights of at-risk maritime workers.

On the other hand, Kalliatakis et al. (2017, 2019) demonstrated the effectiveness of leveraging advanced computer vision techniques to identify human rights violations, including but not limited to child labor, police violence, refugees, child soldiers, forced child marriages, and displacements from images. Such advancements could enhance advocacy efforts by strengthening documentation of abuses, mobilizing timely humanitarian responses, and supporting evidence-based campaigns to protect vulnerable populations and promote accountability for human rights violations in pursuit of social justice.

Remote Sensing and Computer Vision to Assist in Disaster Relief

Satellite imagery provides critical information to monitor disaster events, detect and map affected infrastructure, assess the extent of damage, and identify safe evacuation routes and areas requiring urgent assistance. Therefore, remote sensing and computer vision techniques, including classification, segmentation, and object detection algorithms, could play a crucial role in converting raw satellite imagery into actionable humanitarian intelligence. Such information empowers humanitarian social workers by providing accurate and timely assessments of disaster impacts, enabling targeted interventions, informed evacuation planning, and the efficient allocation of relief resources. For affected populations, this translates into improved safety, timely support, and more effective recovery and reconstruction efforts.

Remote sensing is widely used to support humanitarian relief efforts by providing timely, accurate data for monitoring, assessing, and responding to natural disasters, including floods. For instance, Alizadeh et al. (2022) used advanced computer vision algorithms to estimate floodwater depth by analyzing paired images of submerged stop signs and augmenting flood gauge data. They complemented these data with classical computational algorithms to identify optimized, flood-free evacuation routes. The research incorporated participatory feedback from local residents and rescuers to enhance the practical applicability and effectiveness of evacuation planning. Similarly, Schneibel et al. (2022) demonstrated the use of remote sensing to automate flood monitoring for rapid situational awareness and to provide quick damage assessment of buildings and infrastructure, using satellite data from the Cyclone Idai disaster in Mozambique. By demonstrating high accuracy in flood monitoring (IoU of 80%) and damage assessment for road (IoU of 65.64%) and building detection (F1 score of 71.34%), such algorithms could be confidently deployed to enhance the rapid provision of reliable and accessible crisis information. This could directly support humanitarian relief operations conducted by

NGO's through rapid situation assessments, optimizing the allocation of relief resources and enhancing overall decision-making capabilities.

Similarly, remote sensing can assist in humanitarian relief and human rights monitoring in the aftermath of human-made disasters, such as wars, as exemplified by Mueller et al. (2021), who used advanced computer vision algorithms to identify destroyed buildings from patch-level satellite imagery collected during the Syrian War in Aleppo. By validating the generalizability and robustness of their model against external bombing event data, Mueller et al. (2021) demonstrated that their approach could be reliably deployed in real-world scenarios, thereby offering the potential to support humanitarian efforts through prioritized aid delivery, streamlined resource allocation, and accelerated post-conflict reconstruction planning.

Computer Vision to Assist People with Disability

Computer vision techniques can facilitate solutions to assist individuals with disabilities by enhancing accessibility and inclusion, bridging communication gaps, identifying barriers, and promoting equitable social participation and mobility. The utilization of computer vision to assist deaf and mute individuals through sign language detection—where sign language gestures are automatically recognized and translated into text (Swaroop et al., 2024) or speech (Rajanishree et al., 2022), understandable by individuals unfamiliar with sign language—highlights AI's potential to bridge communication barriers and promote equitable social inclusion. Open-access and deployable tools, such as OpenCV, have enabled seamless integration between gesture recognition algorithms and real-time video processing (Vasanth Kumar & Fawziya, 2024; Swaroop et al., 2024).

In addition, computer vision can assist individuals with disabilities, as demonstrated by Wu et al. (2019), who employed image classification techniques to identify buildings across Milwaukee, Chicago, Los Angeles, and other urban areas based on the presence of handicap-accessible ramps. Such technology could be integrated into future applications to provide real-time, location-based accessibility information, enabling individuals with disabilities to navigate urban environments better.

4.3.4 Virtual Reality and Chatbots to Assist Advocacy Stakeholders

The advanced reasoning capabilities of LLMs and CV have paved the way for generative tools—such as OpenAI's ChatGPT—to develop reliable chatbots and VR experiences. These innovative tools significantly enhance the capacity of advocacy groups and NGOs, effectively supporting advocates and staff through interactive training and educational resources, while simultaneously providing

meaningful assistance and timely guidance to victims and the vulnerable populations they serve.

Virtual Reality and Chatbots Supporting Advocacy Efforts

Virtual Reality and chatbots can assist NGOs in training and recruiting advocates by enabling them to practice advocacy in simulated environments. For instance, Pillis et al. developed the “LGBTQIA+ Advocacy Simulator,” which utilizes AI-generated virtual characters powered by generative conversational AI. Advocates engaged with realistic simulations of workplace interactions centered around scenarios such as conversations about “coming out.” This was accomplished through interactive, web-based technologies that allowed users to either directly control a virtual character’s dialogue (“First Person” mode) or guide character interactions indirectly by instructing AI-generated responses (“Third Person” mode). Results indicated significant improvements in advocates’ empathy and sympathy, with a mean increase of up to 0.73 (on a scale of 7), demonstrating safe and educational practice of advocacy skills in sensitive social contexts.

Similarly, Baugerud et al. developed an AI-based child avatar system designed to help forensic interviewers practice interviewing techniques with simulated child victims of abuse. The system does so through a “virtual child avatar” trained on 741 mock forensic interviews. Interviewers interacted with the child avatar in immersive VR scenarios simulating sexual and physical abuse disclosures, allowing realistic question-and-response exchanges. Results showed that interviewers asked predominantly open-ended questions and rated their interactions positively, finding the avatar’s responses realistic (mean realism rating = 3.44 out of 5) and comfortable engaging with the virtual child (mean comfort rating = 3.59 out of 5). This approach highlights the potential for VR and generative AI to significantly improve training effectiveness and enhance advocates’ practical skills in sensitive interviewing contexts.

Chatbots to Assist Victims and Vulnerable Populations

Robust chatbots have also been developed to assist advocates in supporting victims and vulnerable populations. For example, Bauer et al. (2020) developed a chatbot to help survivors of sexual harassment by directing them to appropriate institutions and increasing documentation of underreported harassment cases. They achieved this by creating a three-component system consisting of (1) harassment type classification, (2) extracting spatiotemporal information (such as locations, organizations, or individuals involved), and (3) managing dialogues with users. The system achieved approximately 80% accuracy in harassment type classification and over 90% in extracting locations and dates, demonstrating its ability to facilitate precise documentation and timely guidance. Such a system could provide meaningful assistance to survivors, empowering those who may be hesitant to disclose

harassment directly to individuals, stepping in to offer immediate support when social workers are not readily available, and swiftly identifying key entities involved to facilitate timely reporting. Consequently, it could significantly enhance social justice and advocacy efforts by increasing awareness, improving documentation accuracy, and accelerating interventions to pursue accountability and protection for victims.

Similarly, Mutinda and Muchiri (2024) developed a multilingual chatbot to support victims of gender-based violence (GBV) in Kenya, designed to provide emotional counseling and immediate referrals to relevant local services. They achieved this by building a chatbot trained on bilingual datasets (English and Kiswahili). The chatbot attained an accuracy of 63%–92%, demonstrating its ability to provide accessible emotional support and tailored referrals in locally prevalent languages, thereby significantly contributing to advocacy efforts and improved outcomes for GBV victims. Specifically, this chatbot could provide crucial support to social workers by extending immediate assistance in rural communities lacking prompt access to services, reducing stigma through anonymity, and systematically documenting cases to strengthen advocacy and social justice initiatives.

Chatbots toward Mental Health Advocacy

The use of chatbots and other AI tools to assist in mental health care is a widely explored topic, as demonstrated by Alathamneh et al. (2024), who developed a chatbot to provide empathetic, contextually relevant conversational support for mental health patients. They accomplished this by training the chatbot on a diverse set of mental health datasets, including mental health counseling conversations, frequently asked questions (FAQs), and anonymized conversational data from various health-care resources (e.g., WebMD, Mayo Clinic, HealthLine). In doing so, the authors hope to equip the chatbot with professional-level mental health guidance and expert-informed advice, thereby significantly improving access to real-time mental health support, notably in cases where professional counseling services (including Licensed Clinical Social Workers) may not be immediately available or individuals who face stigma and prefer seeking initial support anonymously before gaining the confidence to approach a counselor directly.

Similarly, Broadbent et al. (2024) demonstrated the effectiveness of ElliQ, an AI-driven social robot designed to alleviate loneliness among elderly individuals through personalized companionship, proactive engagement, daily conversation prompts, wellness check-ins, and cognitive stimulation activities. The study showed that 80% of participants reported reduced feelings of loneliness and improved emotional well-being, highlighting the potential of AI-driven assistive robots to address mental health concerns among the elderly. Such proactive, automated interactions could substantially relieve social workers in elderly care facilities of routine

responsibilities, enabling them to devote their time and professional expertise to situations requiring specialized intervention.

5 Discussion

This study conducted an extensive review of AI applications toward advocacy and social justice, aiming to assist social workers and professionals from related disciplines in understanding and leveraging AI tools to advance their advocacy objectives and promote social justice. We identified five prevalent areas of AI use in advocacy and social justice: (1) social media mining, (2) computer vision and remote sensing for humanitarian challenges, (3) virtual reality and chatbots for assisting advocacy stakeholders, (4) predictive modeling to address homelessness, and (5) NLP for legal predictions. Through this review, we uncovered key insights into how AI methods are actively employed to promote fairness, enhance advocacy efforts, and support equitable outcomes across various social justice contexts.

The “AI Boom,” which has unfolded since the late 2010s, has invited extensive criticism regarding job displacement, ethical concerns, model explainability and transparency, privacy risks arising from the extensive use of training data, and the potential for misuse. These criticisms have intensified significantly following the emergence of generative AI models such as ChatGPT, further raising concerns about the substantial environmental impacts of the resources required to train these large-scale models, alongside other contentious issues, including copyright infringement related to training data. However, despite these valid concerns, our study illustrates that AI, when thoughtfully designed, responsibly implemented, and deployed with intentions for social good, can significantly benefit humanity by advancing social justice initiatives.

While the use of AI in advocacy and social justice remains relatively modest compared to other prominent AI application fields such as healthcare, e-commerce, and social media, continued innovation and mindful application—coupled with reliable and representative data—offer promising opportunities to enhance equity, protect human rights, and empower communities. To effectively execute this vision, it is critical to promote the open availability of AI models and datasets, encourage active collaboration between AI experts and advocacy stakeholders, and develop safe, scalable, and meaningful solutions explicitly aimed at achieving social good.

5.1 Areas of Success

Many of the aforementioned AI tools have been extensively adopted across distinct facilities serving vulnerable populations (e.g., elder care facilities), integrated into nonprofit or nongovernmental organizations that advocate social justice (e.g., Amnesty International), or made accessible to the public for use by both social workers and vulnerable populations. These success stories highlight the importance and impact of

developing these AI tools beyond academic publications, thereby turning them into actionable goods that can benefit society through advocacy and social justice.

5.1.1 Adoption of AI Tools into Facilities Serving Vulnerable Populations

ElliQ, an AI-driven social robot designed to alleviate loneliness among elderly individuals through personalized companionship, proactive engagement, daily conversation prompts, wellness check-ins, and cognitive stimulation activities (as detailed in the Results section above), has been extensively integrated into multiple elderly care facilities, spanning 15 programs across numerous healthcare organizations in the United States and Canada (Broadbent et al., 2024). Extensive follow-up trials demonstrated its effectiveness in reducing loneliness, improving users' emotional well-being, and reducing social workers' workload in these facilities. This prominent and widespread adoption, supported by evidence beyond algorithmic development, highlights AI's potential to enhance the quality of life for vulnerable populations and support relevant stakeholders when proactively implemented in care environments.

5.1.2 Adoption by Nonprofit Advocacy Organizations Toward the Pursuit of Social Justice

As noted earlier, AI techniques such as social media mining and remote sensing could be effectively leveraged to support humanitarian and advocacy efforts in their pursuit of social justice. When embraced and integrated by nonprofit advocacy organizations, these tools could significantly assist social and humanitarian workers in delivering timely relief while enabling advocates to utilize data-driven insights to raise public awareness of injustices. This is perhaps best exemplified by Amnesty International's digital investigations team, Evidence Lab (<https://citizenevidence.org>), which actively advances digital methods to support human rights among populations subjected to injustice by employing several of the aforementioned tools, including social media mining and remote sensing.

Amnesty International's effectiveness is exemplified by its successful development and utilization of the "Strike Tracker" in Syria, which leveraged remote sensing techniques (as detailed in the earlier Results section) to systematically detect and document civilian building destruction resulting from air strikes during the recently concluded Syrian Civil War. By training computer vision models to automatically identify damaged structures in satellite imagery in real time, these tools were effectively deployed to support humanitarian workers in aid and rescue efforts. Additionally, the project provided compelling AI-driven evidence of the destruction caused by the Assad regime's unjustifiable attacks on civilian populations, strengthening Amnesty's advocacy against human rights violations. This initiative, alongside other AI-driven projects by Amnesty International and similar organizations, highlights AI's considerable potential to advance advocacy and social justice efforts when proactively embraced by nonprofit organizations.

5.1.3 Making Tools Publicly Accessible for Vulnerable Populations

Beyond demonstrating efficacy through academic publication, translating AI-driven advancements into accessible, open-source, and practically usable tools for vulnerable populations is crucial. Such accessibility ensures that the potential of AI extends beyond theoretical results, effectively addressing real-world humanitarian needs by empowering individuals who are directly affected.

A variety of publicly accessible AI tools already exist, spanning multiple domains of social and humanitarian support. Examples include mental health platforms such as Wysa (<https://www.wysa.com/>), Woebot (<https://woebothealth.com/>), and Earkick (<https://earkick.com/>), as well as initiatives designed specifically to assist victims of abuse, such as Chatbot Sophia (<https://springact.org/sophia-chatbot/>) and Hope Chat AI (<https://www.domesticshelters.org/hope-chat-ai>). These tools can be particularly beneficial by offering immediate, anonymous support to vulnerable individuals and providing a safe initial step toward seeking professional help, especially for those who might otherwise be hesitant or fearful to approach traditional services directly (Moore et al., 2025). Other examples include AutoThemeGenerator (Yang et al., 2024a, 2024b), a generative AI-powered tool that assists qualitative researchers in performing thematic analysis more efficiently. Such a tool can help advocacy groups and social researchers rapidly process and interpret large volumes of qualitative data—turning raw testimonies, interview transcripts, or focus group notes into actionable insights—thereby accelerating evidence-based advocacy and enabling faster, better-informed interventions.

The successful deployment and public accessibility of these AI-driven tools have been enabled in significant part by collaborative partnerships between nonprofit organizations and AI researchers. Such collaborations reflect a shared commitment to translating technological advances into practical solutions that directly benefit those in need. Importantly, these tools serve as supplementary resources, providing immediate, accessible support while encouraging users to seek professional intervention when ready (Dakanalis et al., 2024; Moore et al., 2025). As such, they complement rather than replace professional assistance, highlighting AI's potential to enhance existing services rather than substitute critical human expertise (Dakanalis et al., 2024; Moore et al., 2025).

5.2 Areas for Growth

5.2.1 Ensuring AI-Led Advocacy Efforts Are Culturally Sensitive

To effectively utilize AI in advocacy and social justice, it is essential to account for cultural factors in human–computer interaction (HCI) (Shafik, 2025). HCI is an interdisciplinary field concerned with designing, evaluating, and implementing interactive computing systems that are accessible, intuitive, and user-friendly. Successful integration of AI-driven solutions into real-world advocacy contexts

relies on understanding users' cultural preferences, behaviors, and expectations to ensure that solutions are well received, widely adopted, and effective in practice (Madon, 1992).

Cultural values and norms vary significantly across countries and communities, reflecting distinct cultures and shaping perceptions, attitudes, and responses toward AI technologies and associated policies (Alba & Mittal, 2021). Due to this cultural variability, individuals and groups may interpret, accept, or resist AI-led advocacy efforts differently, depending on their cultural backgrounds. To ensure equitable and effective dissemination, it is crucial to incorporate culturally informed approaches within HCI frameworks during AI development and implementation. Integrating cultural considerations can enhance user trust, improve engagement, and ultimately maximize the social impact of AI in advancing advocacy and social justice initiatives across diverse global contexts.

5.2.2 Trickling Efforts to Low-Resource Communities

A persistent challenge in scaling AI-led advocacy efforts to vulnerable populations is the disproportionate concentration of tools, datasets, and research in high-resource languages (e.g., English). In contrast, low-resource languages remain underrepresented in AI development and deployment (Alam et al., 2024). This disparity can be attributed to several factors: first, low-resource languages often suffer from limited availability of high-quality, annotated datasets, making it challenging to train robust and accurate models; second, organizations and communities that speak these languages may lack the financial resources, technical expertise, or institutional support necessary to develop and maintain such systems (Mothe, 2024).

The resulting imbalance risks exacerbating the existing Global North-Global South divide and deepening the broader digital divide, where populations in low-resource settings, who often face heightened vulnerabilities, are the least likely to benefit from cutting-edge AI solutions (Yu et al., 2023). This is especially concerning in contexts where social service capacity is already strained, and human resources for social workers or humanitarian responders are scarce. Addressing this gap requires deliberate strategies to localize AI tools, expand multilingual datasets, and invest in capacity-building initiatives within underserved regions.

5.3 Limitations

5.3.1 Misinformation

Despite the considerable promise of AI for advocacy and social justice, its application poses significant risks when used to leverage large-scale user-generated content from online platforms. Advocacy and social justice groups increasingly rely on AI-driven social media mining and text analysis to rapidly disseminate information,

mobilize support, and shape public discourse. However, this reliance can inadvertently amplify misinformation, particularly when out-of-context or misleading narratives are propagated quickly, causing unintended harm to public credibility and diminishing the legitimacy of advocacy efforts. Such misinformation can ultimately undermine trust and credibility in advocacy messages, hindering effective engagement and meaningful social change.

Moreover, modern-day social media algorithms, designed explicitly to personalize content and enhance user engagement, may inadvertently exacerbate the creation of echo chambers—spaces in which users are predominantly exposed to content aligning closely with their existing beliefs. As highlighted by recent research, such personalized algorithmic filtering can significantly heighten polarization and extremity of viewpoints within social networks (Springsteen et al., 2024). This effect is further intensified when negative stereotypes about out-group members or opposing viewpoints permeate the advocacy-driven content, potentially fostering divisiveness rather than understanding and unity. Consequently, misinformation is further perpetuated as users become increasingly insulated from diverse, fact-based perspectives.

To mitigate these risks, AI advocacy stakeholders must critically and continuously assess information validity, context, and sources using advanced AI-driven approaches, including retrieval-augmented generation (RAG) and context-sensitive language models (Deng et al., 2025). These methodologies offer promising avenues for rapidly verifying the accuracy and contextual relevance of information, detecting inconsistencies, and providing balanced perspectives from reliable sources. Such proactive measures can help advocacy groups maintain trust, reduce misinformation, and effectively promote exposure to diverse, credible, and constructive viewpoints.

5.3.2 Fairness and Bias

Another critical limitation concerns the fairness and bias inherent in AI models, arising primarily from historical biases, racism, and systemic inequalities deeply embedded in human-generated data. When AI systems are trained on such historical data, they risk amplifying existing prejudices and discriminatory practices, creating “algorithms of oppression” (Noble, 2018) that systematically disadvantage marginalized or minority communities. A prominent example is the controversy surrounding the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) software (Brennan & Dieterich, 2018), an AI tool used to assess potential recidivism risk. In a widely publicized case, 41-year-old African American Prater, convicted of shoplifting \$86.35 worth of tools, was predicted to have a high risk of future offenses. In contrast, Borden, a white defendant convicted of armed robbery and attempted armed robbery, was predicted as low risk (Angwin et al., 2016).

Contrary to these predictions, Prater committed no subsequent offenses, while Borden is serving an eight-year prison term for later crimes. A follow-up ProPublica study (Angwin et al., 2016) further highlighted that Black defendants were nearly

twice as likely as white defendants to be labeled high risk but not actually re-offend. This exemplifies how biases embedded within historical data can perpetuate unjust outcomes when integrated into predictive modeling.

Henceforth, to counteract such biases, AI practitioners must incorporate participatory approaches and critical frameworks into model development, actively engage diverse communities in the design process, and rigorously examine underlying assumptions to create equitable and socially just AI solutions. Additionally, practitioners can leverage Fair ML solutions such as Fairlearn (Bird et al., 2020) and IBM's AI Fairness 360 (Bellamy et al., 2019). These tools enable users to systematically detect, quantify, and mitigate biases through methods such as fairness-aware optimization, bias metrics evaluation, and explainability analyses, thereby promoting equitable outcomes and transparency in AI-driven decision-making.

5.3.3 AI for Social Harm: Using AI to Target Marginalized Individuals

While AI technologies have significant potential for positive social impact, they can also be employed in ways that harm vulnerable or marginalized individuals. Surveillance tools, such as facial recognition and predictive policing algorithms, can be used to disproportionately monitor, track, and discriminate against minority communities, facilitating oppressive practices or serving personal and political agendas. For example, recent reports suggest that the Trump administration has employed AI-driven surveillance to monitor social media activity of foreign nationals, identifying individuals perceived as supporting terror groups such as Hamas, thereby posing significant threats to civil liberties and freedom of expression (Monahan, 2025).

These applications illustrate the critical importance of teaching ethics and social responsibility to AI practitioners and data scientists. Professionals developing AI tools, as well as advocacy and social justice stakeholders who could potentially use these tools, must remain vigilant, foster critical thinking about the contexts in which their innovations are deployed, and ensure transparency in their applications. To prevent misuse and protect human rights, cross-disciplinary collaboration and community-based approaches among AI experts, policymakers, social advocates, and affected communities are essential. Only through rigorous ethical education, active engagement, and mutual accountability can we ensure that AI systems are developed and used in ways that promote social justice rather than undermine it.

6 Future Directions and Challenges

6.1 Open Sourcing Data

As demonstrated by our results, high-quality, representative, and globally diverse data are essential to achieving tangible outcomes in advocacy and social justice applications of AI. Promoting open science principles, particularly the widespread

sharing and open sourcing of data, will be instrumental in facilitating collaborative research, innovation, and equitable AI model development across international contexts. However, recent trends toward monetizing previously free data, such as API access from popular social media platforms like X (formerly Twitter) (Center of an Informed Public, 2023), present significant barriers. Such monetization disproportionately impacts researchers from under-resourced countries and institutions, exacerbating global disparities by limiting their access to essential data for advocacy and social justice research. Advocating for policies that encourage open data sharing, providing grants to enable equitable data access, and developing international collaborative frameworks will therefore be critical to ensuring inclusive participation and advancing socially beneficial AI research.

6.2 Implementing Models to Enact Social Good

Successfully translating predictive AI models and analytical insights from academic settings into tangible improvements in people's lives requires active collaboration with authorities, advocacy organizations, community representatives, and other relevant stakeholders. Such multi-stakeholder engagement ensures that AI interventions are aligned with real-world needs, can be feasibly implemented, and are accepted by the communities they aim to benefit. In addition, effective deployment of these AI-driven advocacy solutions demands rigorous attention to human-computer interaction (HCI) research. Integrating user-centered design principles (Gulliksen et al., 2003), behavioral and decision-making concepts (Alba et al., 2024), and cultural dimensions of technology (Shafik, 2025) when implementing these AI models through HCI studies is essential to ensure that solutions are accessible, intuitive, culturally sensitive, and genuinely helpful to the diverse populations they seek to empower.

Additionally, to robustly assess the effectiveness and social impact of AI-driven advocacy tools, empirical validation through randomized controlled trials (RCTs) or rigorous observational studies must become standard practice. Such studies will help systematically assess whether AI interventions produce meaningful, real-world improvements in advocacy and social justice outcomes, while simultaneously identifying areas that need further refinement. Ultimately, rigorous testing and iterative evaluation will be critical to turning theoretical AI advances into scalable and impactful social justice interventions.

6.3 Sustainable and Cost-Effective Solutions

A critical challenge facing AI-driven advocacy and social justice initiatives is ensuring sustainability and affordability. Many contemporary generative AI models, particularly large-scale models, have significant environmental footprints due to the

vast computational resources required for training and deployment (Berthelot et al., 2024), including high GPU costs and substantial energy consumption. This presents a paradox: leveraging AI for social good may inadvertently exacerbate environmental harm and widen existing digital divides, especially disadvantaging resource-scarce countries or communities lacking necessary infrastructure. Addressing this paradox requires a shift toward more sustainable AI practices, such as developing smaller yet robust models (Schick & Schütze, 2021) or employing resource-efficient techniques that enable these large AI models to be trained in an environmentally and cost-efficient manner (Han et al., 2024). These strategies not only reduce computational demands and environmental impact but also promote greater accessibility and practical implementation across diverse global settings, aligning AI-driven advocacy more closely with principles of equity and environmental responsibility.

References

- Alam, F., Chowdhury, S. A., Boughorbel, S., & Hasanain, M. (2024, March). LLMs for low resource languages in multilingual, multimodal and dialectal settings. In *Proceedings of the 18th conference of the European chapter of the association for computational linguistics: tutorial abstracts*, pp. 27–33. <https://doi.org/10.18653/v1/2024.eacl-tutorials.5>
- Alathamneh, R., Al Ameri, S., & Al Obeidat, F. (2024). Artificial intelligence in mental health care: The t5 chatbot project. In *2024 11th International Conference on Social Networks Analysis, Management and Security (SNAMS)*, pp. 212–218. <https://doi.org/10.1109/SNAMS64316.2024.10883804>
- Alba, C., & An, R. (2023). Using mobile phone data to assess socio-economic disparities in unhealthy food reliance during the COVID-19 pandemic. *Health Data Science*, 3, 0101. <https://doi.org/10.34133/hds.0101>
- Alba, C., & Mittal, M. M. (2021). Sociocultural behavioral traits in modelling the prediction of COVID-19 infection rates. *Journal of Humanities and Applied Social Sciences*, 3(5), 339–355. <https://doi.org/10.1108/JHASS-07-2021-0128>
- Alba, C., Walasek, L., & Spektor, M. S. (2024). Attention-driven imitation in consumer reviews. *Decision*. <https://doi.org/10.1037/dec0000238>
- Alizadeh, B., Li, D., Hillin, J., Meyer, M. A., Thompson, C. M., Zhang, Z., & Behzadan, A. H. (2022). Human-centered flood mapping and intelligent routing through augmenting flood gauge data with crowdsourced street photos. *Advanced Engineering Informatics*, 54, 101730. <https://doi.org/10.1016/j.aei.2022.101730>
- Anandhi, R., Devi, V. A., Devi, B. K., Kavin, B. P., & Seng, G. H. (2024). Croa-based feature selection with bert model for detecting the offensive speech in twitter data. *Journal of Autonomous Intelligence*, 7(3), 1122. <https://doi.org/10.32629/jai.v7i3.1122>
- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). *Machine bias*. Propublica. Retrieved April 19, 2019, from <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- Anokwa, Y., Hartung, C., Brunette, W., Borriello, G., & Lerer, A. (2009). Open source data collection in the developing world. *Computer*, 42(10), 97–99. <https://doi.org/10.1109/MC.2009.328>
- Barta, T., & Kheng, J. H. (1997). A knowledge-based housing discrimination advisory system. *New Review of Applied Expert Systems*, 3, 97–108.
- Bauer, T., Devrim, E., Glazunov, M., Jaramillo, W. L., Mohan, B., & Spanakis, G. (2020). #metoomaasticht: Building a chatbot to assist survivors of sexual harassment. In *Machine learning and knowledge discovery in databases: International workshops of ECML PKDD*

- 2019, Würzburg, Germany, September 16–20, 2019, proceedings, Part I, pp. 503–521. https://doi.org/10.1007/978-3-030-43823-4_41
- Bellamy, R. K., Dey, K., Hind, M., Hoffman, S. C., Houde, S., Kannan, K., Lohia, P., Martino, J., Mehta, S., Mojsilović, A., et al. (2019). Ai fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. *IBM Journal of Research and Development*, 63(4/5), 4–1. <https://doi.org/10.1147/JRD.2019.2942287>
- Berthelot, A., Caron, E., Jay, M., & Lefèvre, L. (2024). Estimating the environmental impact of generative-AI services using an lca-based methodology. *Procedia CIRP*, 122, 707–712. <https://doi.org/10.1016/j.procir.2024.01.098>
- Bird, S., Dudík, M., Edgar, R., Horn, B., Lutz, R., Milan, V., Sameki, M., Wallach, H., & Walker, K. (2020). *Fairlearn: A toolkit for assessing and improving fairness in AI*. Microsoft, Technical Report MSR-TR-2020-32. <https://www.jmlr.org/papers/volume24/23-0389/23-0389.pdf>
- Bozhidarova, M., Chang, J., Ale-Rasool, A., Liu, Y., Ma, C., Bertozzi, A. L., Brantingham, P. J., Lin, J., & Krishnagopal, S. (2023). Hate speech and hate crimes: A data-driven study of evolving discourse around marginalized groups. In *2023 IEEE International Conference on Big Data (BigData)*, pp. 3107–3116. <https://doi.org/10.1109/BigData59044.2023.10386312>
- Braun, A., Fakhri, F., & Hochschild, V. (2019). Refugee camp monitoring and environmental change assessment of Kutupalong, Bangladesh, based on radar imagery of sentinel-1 and alos-2. *Remote Sensing*, 11(17), 2047. <https://doi.org/10.3390/rs11172047>
- Brennan, T., & Dieterich, W. (2018). Correctional offender management profiles for alternative sanctions (compas). In *Handbook of recidivism risk/needs assessment tools* (pp. 49–75). Wiley. <https://doi.org/10.1002/9781119184256.ch3>
- Broadbent, E., Loveys, K., Ilan, G., Chen, G., Chilukuri, M., Boardman, S., Doraiswamy, P., & Skuler, D. (2024). ElliQ, an AI-driven social robot to alleviate loneliness: Progress and lessons learned. *The Journal of Aging Research & Lifestyle*, 13, 22–28. <https://doi.org/10.14283/jarlife.2024.2>
- Center of an Informed Public. (2023). *Twitter's API access changes could mark 'end of an era' in academic research on the platform*. <https://www.cip.uw.edu/2023/02/02/twitters-api-access-changes-academic-research/>
- Chanda, S. S., & Banerjee, D. N. (2024). Omission and commission errors underlying AI failures. *AI & Society*, 39(3), 937–960. <https://doi.org/10.1007/s00146-022-01585-x>
- Chen, Y. D., Abbasi, A., & Chen, H. (2008). Developing ideological networks using social network analysis and writeprints: A case study of the international Falun Gong movement. In *2008 IEEE International Conference on Intelligence and Security Informatics* (pp. 7–12). IEEE. <https://doi.org/10.1109/ISI.2008.4565021>
- Chit, K. M. M., Shein, Y. Y. C. M. W., Yan, W., & Khine, A. H. (2022). Siren! detecting Burmese hate speech comments on social media. In *2022 14th International Conference on Knowledge and Smart Technology (KST)*, pp. 119–124. <https://doi.org/10.1109/KST53302.2022.9729075>
- Collenette, J., Atkinson, K., & Bench-Capon, T. (2023). Explainable AI tools for legal reasoning about cases: A study on the European court of human rights. *Artificial Intelligence*, 317, 103861. <https://doi.org/10.1016/j.artint.2023.103861>
- Dakanalis, A., Wiederhold, B. K., & Riva, G. (2024). Artificial intelligence: A game-changer for mental health care. *Cyberpsychology, Behavior, and Social Networking*, 27(2), 100–104. <https://doi.org/10.1089/cyber.2023.0723>
- Dalrymple, J., & Boylan, J. (2013). *Effective advocacy in social work*. Sage <https://uk.sagepub.com/en-gb/eur/effective-advocacy-in-social-work/book236728>
- Deng, B., Wang, W., Zhu, F., Wang, Q., & Feng, F. (2025). Cram: Credibility-aware attention modification in LLMs for combating misinformation in rag. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(22), 23760–23768. <https://doi.org/10.1609/aaai.v39i22.34547>
- Desai, M., & Solas, J. (2012). Poverty, development and social justice. In *The SAGE handbook of international social work* (pp. 85–99). Sage. <https://doi.org/10.4135/9781446247594.n5>
- Dupré, D. T. (2023). Explainable answer set programming for legal decision support. *CILC*. <https://doi.org/10.3233/FAIA230994>

- Fu, T., & Chen, H. (2008, June). Analysis of cyberactivism: A case study of online free Tibet activities. In *2008 IEEE International Conference on Intelligence and Security Informatics* (pp. 1–6). IEEE. <https://doi.org/10.1109/ISI.2008.4565020>
- Gokhale, R., & Fasli, M. (2017). Deploying a co-training algorithm to classify human-rights abuses. In *2017 International conference on the frontiers and advances in data science (FADS)*, pp. 108–113. <https://doi.org/10.1109/FADS.2017.8253206>
- Gokhale, R., & Fasli, M. (2018). Matrix factorization for co-training algorithm to classify human rights abuses. In *2018 IEEE international conference on big data (big data)*, pp. 2170–2179. <https://doi.org/10.1109/BigData.2018.8622397>
- Gulliksen, J., Göransson, B., Boivie, I., Blomkvist, S., Persson, J., & Cajander, Å. (2003). Key principles for user-centred systems design. *Behaviour and Information Technology*, 22(6), 397–409. <https://doi.org/10.1080/01449290310001624329>
- Gupta, V., Ponsam, J. G., & Vangavarapu, C. (2023). Racial slur detection using natural language processes. In *2023 International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering (ICECONF)*, pp. 1–9. <https://doi.org/10.1109/ICECONF57129.2023.10084027>
- Gustafsson, P. U., Lachmann, T., & Laukka, P. (2025). Machine learning predicts accuracy in eyewitnesses' voices. *Journal of Nonverbal Behavior*, 49(1), 155–169. <https://doi.org/10.1007/s10919-024-00474-9>
- Habernal, I., Faber, D., Recchia, N., Bretthauer, S., Gurevych, I., Spiecker genannt Döhmann, I., & Burchard, C. (2024). Mining legal arguments in court decisions. *Artificial Intelligence and Law*, 32(3), 1–38. <https://doi.org/10.1007/s10506-023-09361-y>
- Haimson, O. L. (2019). Mapping gender transition sentiment patterns via social media data: Toward decreasing transgender mental health disparities. *Journal of the American Medical Informatics Association*, 26(8–9), 749–758. <https://doi.org/10.1093/jamia/ocz056>
- Han, Z., Gao, C., Liu, J., Zhang, J., & Zhang, S. Q. (2024). Parameter-efficient fine-tuning for large models: A comprehensive survey. *Transactions on Machine Learning Research*. <https://openreview.net/pdf?id=HsCS8b6zj>
- Hofmann, V., Kalluri, P. R., Jurafsky, D., & King, S. (2024). AI generates covertly racist decisions about people based on their dialect. *Nature*, 633(8028), 147–154. <https://doi.org/10.1038/s41586-024-07856-5>
- Hüsünbeyi, Z. M., Akar, D., & Özgür, A. (2022). Identifying hate speech using neural networks and discourse analysis techniques. In *Proceedings of the first workshop on language technology and resources for a fair, inclusive, and safe society within the 13th language resources and evaluation conference*, pp. 32–41. <https://aclanthology.org/2022.lateraisse-1.5/>
- Imran, A. S., Hodnefeld, H., Kastrati, Z., Fatima, N., Daudpota, S. M., & Wani, M. A. (2023). Classifying European court of human rights cases using transformer-based techniques. *IEEE Access*, 11, 55664–55676. <https://doi.org/10.1109/ACCESS.2023.3279034>
- Jachens, L., Houdmont, J., & Thomas, R. (2019). Effort–reward imbalance and burnout among humanitarian aid workers. *Disasters*, 43(1), 67–87. <https://doi.org/10.1111/disa.12288>
- Jaidka, K., Chen, T., Chesterman, S., Hsu, W., Kan, M.-Y., Kankanhalli, M., Lee, M. L., Seres, G., Sim, T., Taeihagh, A., et al. (2025). Misinformation, disinformation, and generative AI: Implications for perception and policy. *Digital Government: Research and Practice*, 6(1), 1–15. <https://doi.org/10.1145/3689372>
- Jeon, S.-W. (2020). A study on the social phenomenon and perception of sports sexual violence through big data analysis of social media. *Ethiopian Journal of Health Development*, 34(3) <https://www.ajol.info/index.php/ejhd/article/view/198772>
- Jimenez, J., Pasztor, E. M., Chambers, R. M., & Fujii, C. P. (2014). *Social policy and social change: Toward the creation of social and economic justice*. Sage. <https://study.sagepub.com/jimenez2e>
- Kalliatakis, G., Ehsan, S., Fasli, M., Leonardis, A., Gall, J., & McDonald-Maier, K. D. (2017). Detection of human rights violations in images: Can convolutional neural networks help?

- International Conference on Computer Vision Theory and Applications*, 6, 289–296. <https://doi.org/10.5220/0006133902890296>
- Kalliatakis, G., Ehsan, S., Leonardis, A., Fasli, M., & McDonald-Maier, K. D. (2019). Exploring object-centric and scene-centric cnn features and their complementarity for human rights violations recognition in images. *IEEE Access*, 7, 10045–10056. <https://doi.org/10.1109/ACCESS.2019.2891745>
- Kaur, A., & Bozic, B. (2019). Convolutional neural network-based automatic prediction of judgments of the European court of human rights. *AICS*, 458–469. https://ceur-ws.org/Vol-2563/aics_42.pdf
- Khatua, A., Cambria, E., & Khatua, A. (2018). Sounds of silence breakers: Exploring sexual violence on twitter. In *2018 IEEE/ACM international conference on advances in social networks analysis and mining (ASONAM)*, pp. 397–400. <https://doi.org/10.1109/ASONAM.2018.8508576>
- Kihlman, R., & Fasli, M. (2019). Augmenting co-training with recommendations to classify human rights violations. In *2019 IEEE International Conference on Big Data (Big Data)*, pp. 2299–2308. <https://doi.org/10.1109/BigData47090.2019.9005478>
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444. <https://doi.org/10.1038/nature14539>
- Lee, C. S., & Jang, A. (2023). Questing for justice on twitter: Topic modeling of# stopasianhate discourses in the wake of Atlanta shooting. *Crime & Delinquency*, 69(13–14), 2874–2900. <https://doi.org/10.1177/00111287211057855>
- Li, K. (2022). An evaluation of automation on misogyny identification (AMI) and deep-learning approaches for hate speech-highlight on graph convolutional networks and neural networks. In *2022 International Conference on Computers, Information Processing and Advanced Education (CIPAE)*, pp. 239–244. <https://doi.org/10.1109/CIPAE55637.2022.00058>
- Lloyd, C., King, R., & Chenoweth, L. (2002). Social work, stress and burnout: A review. *Journal of Mental Health*, 11(3), 255–265. <https://doi.org/10.1080/09638230020023642>
- Madon, S. (1992). Computer-based information systems for development planning: The significance of cultural factors. *The Journal of Strategic Information Systems*, 1(5), 250–257. [https://doi.org/10.1016/0963-8687\(92\)90014-N](https://doi.org/10.1016/0963-8687(92)90014-N)
- McCarthy, J., Rochester, N., & Shannon, C. (1956). *Dartmouth workshop*. http://aigraph.csl.tu.org/courses/06/Dartmouth_workshop.pdf
- McDonald, G. G., Costello, C., Bone, J., Cabral, R. B., Farabee, V., Hochberg, T., Kroodsmas, D., Mangin, T., Meng, K. C., & Zahn, O. (2021). Satellites can reveal global extent of forced labor in the world's fishing fleet. *Proceedings of the National Academy of Sciences*, 118(3), e2016238117. <https://doi.org/10.1073/pnas.2016238117>
- Miller, D. (1978). Democracy and social justice. *British Journal of Political Science*, 8(1), 1–19. <https://doi.org/10.1017/S0007123400001198>
- Mochales-Palau, R., & Moens, M. F. (2007). Study on sentence relations in the automatic detection of argumentation in legal cases. In *Proceedings of the 2007 conference on Legal Knowledge and Information Systems: JURIX 2007: The Twentieth Annual Conference*, pp. 89–98. <https://dl.acm.org/doi/10.5555/1565610.1565624>
- Monahan, T. (2025). *Surveillance in Trump's America*. <https://doi.org/10.24908/ss.v23i1.19220>
- Moore, B., Herington, J., & Tekin, Ş. (2025). The integration of artificial intelligence-powered psychotherapy Chatbots in pediatric care: Scaffold or substitute? *The Journal of Pediatrics*, 280. <https://doi.org/10.1016/j.jpeds.2025.114509>
- Moser, E., Meyer, S., Schmidhuber, M., Ketterer, D., & Eberhardt, M. (2022). Argo: Towards small vessel detection for humanitarian purposes. *IJCAI*, 2022, 5245–5248. <https://doi.org/10.24963/ijcai.2022/728>
- Mothe, J. (2024). Shaping the future of endangered and low-resource languages---Our role in the age of LLMs: A keynote at ECIR 2024. In *ACM SIGIR Forum* (Vol. 58, pp. 1–13). ACM. <https://doi.org/10.1145/3687273.3687280>

- Mueller, H., Groeger, A., Hersh, J., Matranga, A., & Serrat, J. (2021). Monitoring war destruction from space using machine learning. *Proceedings of the National Academy of Sciences*, 118(23), e2025400118. <https://doi.org/10.1073/pnas.2025400118>
- Mumford, J., Atkinson, K., & Bench-Capon, T. (2023). Combining a legal knowledge model with machine learning for reasoning with legal cases. In *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, pp. 167–176. <https://doi.org/10.1145/3594536.3595158>
- Mumford, J., Atkinson, K., & Bench-Capon, T. (2024). Annotated insights into legal reasoning: A dataset of article 6 echr cases. *Argument & Computation*, 15(2), 113–119. <https://doi.org/10.3233/AAC-240002>
- Mutinda, S., & Muchiri, H. (2024). A bi-lingual counselling chatbot application for support of gender based violence victims in Kenya. In *2024 5th International Conference on Smart Sensors and Application (ICSSA)*, pp. 1–6. <https://doi.org/10.1109/ICSSA62312.2024.10788649>
- Noble, S. U. (2018). Algorithms of oppression: How search engines reinforce racism. In *Algorithms of oppression*. New York University Press. <https://doi.org/10.2307/j.ctt1pw9w5>
- Pillis, D., Pataranutaporn, P., Maes, P., & Sra, M. (2024, March). AI comes out of the closet: Using AI-generated virtual characters to help individuals practice LGBTQIA+ advocacy. In *Proceedings of the 29th International Conference on Intelligent User Interfaces*, pp. 686–698. <https://doi.org/10.1145/3640543.3645213>
- Poudyal, P., Gonçalves, T., & Quaresma, P. (2019). Using clustering techniques to identify arguments in legal documents. *ASAIL@ ICAIL*, 2385. <https://ceur-ws.org/Vol-2385/paper2.pdf>
- Purao, S., Garfield, M., Gu, X., & Bhetwal, P. (2019). Predicting the slide to long-term homelessness: Model and validation. In *2019 IEEE 21st Conference on Business Informatics (CBI)*, Vol. 1, pp. 31–40. <https://doi.org/10.1109/CBI.2019.00011>
- Quemy, A., & Wrembel, R. (2020). On integrating and classifying legal text documents. In *International Conference on Database and Expert Systems Applications*, pp. 385–399. https://doi.org/10.1007/978-3-030-59003-1_25
- Quinn, J. A., Nyhan, M. M., Navarro, C., Coluccia, D., Bromley, L., & Luengo-Oroz, M. (2018). Humanitarian applications of machine learning with remote-sensing data: Review and case study in refugee settlement mapping. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2128), 20170363. <https://doi.org/10.1098/rsta.2017.0363>
- Rajanishree, M., Ahmed, N. N., Panchani, Y., Aravindan, S., & Jadhav, V. (2022). Sign language conversion to speech with the application of knn algorithm. In *2022 Sixth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC)*, pp. 886–890. <https://doi.org/10.1109/I-SMAC55078.2022.9987421>
- Rathod, R. G., Barve, Y., Saini, J. R., & Rathod, S. (2023). From data pre-processing to hate speech detection: An interdisciplinary study on women-targeted online abuse. In *2023 3rd International conference on intelligent technologies (CONIT)*, pp. 1–8. <https://doi.org/10.1109/CONIT59222.2023.10205571>
- Redmond, J. (2021). Moderate resolution remote sensing and machine learning for human rights monitoring: The case of Rakhine State, Myanmar. In *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, pp. 2317–2320. <https://doi.org/10.1109/IGARSS47720.2021.9553953>
- Russell, S. J., & Norvig, P. (2016). *Artificial intelligence: A modern approach*. Pearson.
- Sabatovych, I. (2019). Do social media create revolutions? using twitter sentiment analysis for predicting the maidan revolution in Ukraine. *Global Media and Communication*, 15(3), 275–283. <https://doi.org/10.1177/1742766519872780>
- Sanghi, A., Sinha, A., Venkatesh, A., & Das, B. (2021). A framework to capture the shift in dynamics of a multi-phase protest—a case study of Hong Kong protests. *Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2020*, 3, 95–110. https://doi.org/10.1007/978-981-15-9774-9_10

- Santosh, T., Chowdhury, I., Xu, S., & Grabmair, M. (2024). The craft of selective prediction: Towards reliable case outcome classification - An empirical study on European court of human rights cases. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 3656–3674. <https://doi.org/10.18653/v1/2024.findings-emnlp.208>
- Sari, I., Kosasih, R., & Fahrurrozi, A. (2022). Implementation of Machine Learning in Predicting Length of Punishment at Bandung Court. In *2022 International Conference of Science and Information Technology in Smart Administration (ICSINTESA)* (pp. 12–17). IEEE. <https://doi.org/10.1109/ICSINTESA56431.2022.10041693>
- Sawhney, R., Mathur, P., Jain, T., Gautam, A. K., & Shah, R. (2021). Multitask learning for emotionally analyzing sexual abuse disclosures. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4881–4892. <https://doi.org/10.18653/v1/2021.naacl-main.387>
- Schick, T., & Schütze, H. (2021). It's not just size that matters: Small language models are also few-shot learners. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2339–2352. <https://doi.org/10.18653/v1/2021.naacl-main.185>
- Schirmer, M., Brechenmacher, C., Jashari, E., & Pfeffer, J. (2024, May). GENTRAC: A tool for tracing trauma in genocide and mass atrocity court transcripts.
- Schneibel, A., Merkle, N., Wieland, M., Lechner, K., Azimi, S., Henkel, F., Henry, C., Kiefl, R., Yuan, X., & Gaehler, M. (2022). User-driven flood response & monitoring information—key findings of the data 4human project. In *2022 IEEE Global Humanitarian Technology Conference (GHTC)*, pp. 46–53. <https://doi.org/10.1109/GHTC55712.2022.9911021>
- Selvam, V. N. (2024). “Empowering communities”: Harnessing energy for human rights and social justice. *International Journal of Law Management & Humanities*, 7(1), 353. <https://doi.org/10.1000/IJLMH.116705>
- Shafik, W. (2025). Human-computer interaction (HCI) technologies in socially-enabled artificial intelligence. In *Future of digital technology and AI in social sectors* (pp. 121–150). IGI Global. <https://doi.org/10.4018/979-8-3693-5533-6.ch005>
- Sivasankari, S., & Vadivu, G. (2022). Tracing the fake news propagation path using social network analysis. *Soft Computing*, 26(23), 12883–12891. <https://doi.org/10.1007/s00500-021-06043-2>
- Springsteen, J., Yeoh, W., & Christenson, D. (2024). Algorithmic filtering, out-group stereotype, and polarization on social media. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, pp. 1782–1790. <https://doi.org/10.5555/3635637.3663040>
- Stuart, F., Riley, A., & Pourreza, H. (2020). A human-machine partnered approach for identifying social media signals of elevated traumatic grief in Chicago gang territories. *PLoS One*, 15(7), e0236625. <https://doi.org/10.1371/journal.pone.0236625>
- Subramani, S., Michalska, S., Wang, H., Du, J., Zhang, Y., & Shakeel, H. (2019). Deep learning for multi-class identification from domestic violence online posts. *IEEE Access*, 7, 46210–46224. <https://doi.org/10.1109/ACCESS.2019.2908827>
- Suk, J., Abhishek, A., Zhang, Y., Ahn, S. Y., Correa, T., Garlough, C., & Shah, D. V. (2021). #metoo, networked acknowledgment, and connective action: How “empowerment through empathy” launched a social movement. *Social Science Computer Review*, 39(2), 276–294. <https://doi.org/10.1177/0894439319864882>
- Swanson, K., Ravi, A., Saleh, S., Weia, B., Pleasants, E., & Arvisais-Anhalt, S. (2023). Effect of recent abortion legislation on twitter user engagement, sentiment, and expressions of trust in clinicians and privacy of health information: Content analysis. *Journal of Medical Internet Research*, 25, e46655. <https://doi.org/10.2196/46655>
- Swaroop, S., Prasad, K. Y., Reddy, G. S. S., Rachavelpula, M., & Jogalekar, U. A. (2024). Real time sign language detection. In *2024 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pp. 1–7. <https://doi.org/10.1109/ACDSA59508.2024.10467736>

- Tong, X., Li, Y., Li, J., Bei, R., & Zhang, L. (2022). What are people talking about in #backlives-matter and #stopasianhate? Exploring and categorizing twitter topics emerged in online social movements through the latent Dirichlet allocation model. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 723–738. <https://doi.org/10.1145/3514094.3534202>
- Tsai, F.-J., Huang, W.-L., & Chan, C.-C. (2009). Occupational stress and burnout of lawyers. *Journal of Occupational Health*, 51(5), 443–450. <https://doi.org/10.1539/joh.18179>
- Umel, A. (2024). Facebook and social representations of Filipino migrant life in Germany: A reflexive computational approach. *Frontiers in Human Dynamics*, 5, 1284711. <https://doi.org/10.3389/fhumd.2023.1284711>
- Vasanth Kumar C., & Fawziya, A. (2024). Ai-enhanced sign language interpreter. In *International Conference on Computer, Communication, and Signal Processing*, pp. 186–198. https://doi.org/10.1007/978-3-031-73617-9_15
- Venkatesan, S., Valecha, R., Yaraghi, N., Oh, O., & Rao, H. R. (2021). Influence in social media: An investigation of tweets spanning the 2011 Egyptian revolution. *MIS Quarterly*, 45(4). <https://doi.org/10.25300/MISQ/2021/15297>
- Wahyuningroem, S. L., Sirait, R., Uljanatunnisa, U., & Heryadi, D. (2024). Youth political participation and digital movement in Indonesia: The case of #reformasidikorupsi and #tolakomnibuslaw. *F1000Research*, 12, 543. <https://doi.org/10.12688/f1000research.122669.3>
- Wang, J., Lan, C., Liu, C., Ouyang, Y., Qin, T., Lu, W., Chen, Y., Zeng, W., & Yu, P. S. (2022). Generalizing to unseen domains: A survey on domain generalization. *IEEE Transactions on Knowledge and Data Engineering*, 35(8), 8052–8072. <https://doi.org/10.1109/TKDE.2022.3178128>
- Wang, X., Zhang, X., Hoo, V., Shao, Z., & Zhang, X. (2024). Legalreasoner: A multi-stage framework for legal judgment prediction via large language models and knowledge integration. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3496666>
- Watts, L., & Hodgson, D. (2019). *Social justice theory and practice for social work*. Springer. <https://doi.org/10.1007/978-981-13-3621-8>
- Wu, J., Hu, W., Coelho, J., Nitu, P., Paul, H. R., Madiraju, P., Smith, R. O., & Ahamed, S. I. (2019). Identifying buildings with ramp entrances using convolutional neural networks. In *2019 IEEE 43rd Annual Computer Software and Applications Conference (COMPSAC)*, Vol. 2, pp. 74–79. <https://doi.org/10.1109/COMPSAC.2019.10186>
- Xu, S., Tyss, S., Ichim, O., Plank, B., & Grabmair, M. (2024). Through the lens of split vote: Exploring disagreement, difficulty and calibration in legal case outcome classification. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 199–216. <https://doi.org/10.18653/v1/2024.acl-long.13>
- Yang, B., Chen, G., & Luo, X. (2024a). A legal judgment prediction model based on bert, attention, and graph convolutional network. In *2024 IEEE 36th International Conference on Tools with Artificial Intelligence (ICTAI)*, pp. 303–310. <https://doi.org/10.1109/ICTAI62512.2024.00052>
- Yang, Y., Alba, C., Wang, C., Wang, X., Anderson, J., & An, R. (2024b). GPT models can perform thematic analysis in public health studies, akin to qualitative researchers. *Journal of Social Computing*, 5(4), 293–312. <https://doi.org/10.23919/JSC.2024.0024>
- Yildirim, E., Sert, M. F., Kartal, B., & Çalış, Ş. (2023). Non-compliance of the European court of human rights decisions: A machine learning analysis. *International Review of Law and Economics*, 76, 106167. <https://doi.org/10.1016/j.irle.2023.106167>
- Yu, D., Rosenfeld, H., & Gupta, A. (2023). The ‘AI divide’ between the global north and global south. In *World Economic Forum* (Vol. 16). <https://www.weforum.org/stories/2023/01/davos23-ai-divide-global-north-global-south/>

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